

Video Anomaly Detection

Problem Statement

How to detect video anomaly from videos using Vision based techniques ?

Objective

Problem & Safety Risks

Manual CCTV review is expensive and fatigue-prone, with one analyst monitoring eight cameras.

Missed anomalous events create significant safety and liability risks.

Core Objective & Use-Cases

Flag anomalous frames in near-real-time to reduce analyst workload by approximately 80%.

Key use-cases: crowd surges, foreign object intrusion, loitering, and unattended bags.

78%

Review Load Reduction

Substantial 78% reduction in overall analyst review load

< 2s

Alert Latency

Mean time to alert under two seconds for rapid response

< 8%

False Alarm Rate

Maintains a false alarm rate strictly below 8%

< 5ms

Inference / Frame

Inference cost under 5 milliseconds per frame

0.6428

AUC-ROC Score

Benchmark AUC-ROC score achieved across evaluation sets

UCSD Pedestrian Dataset (Ped2)

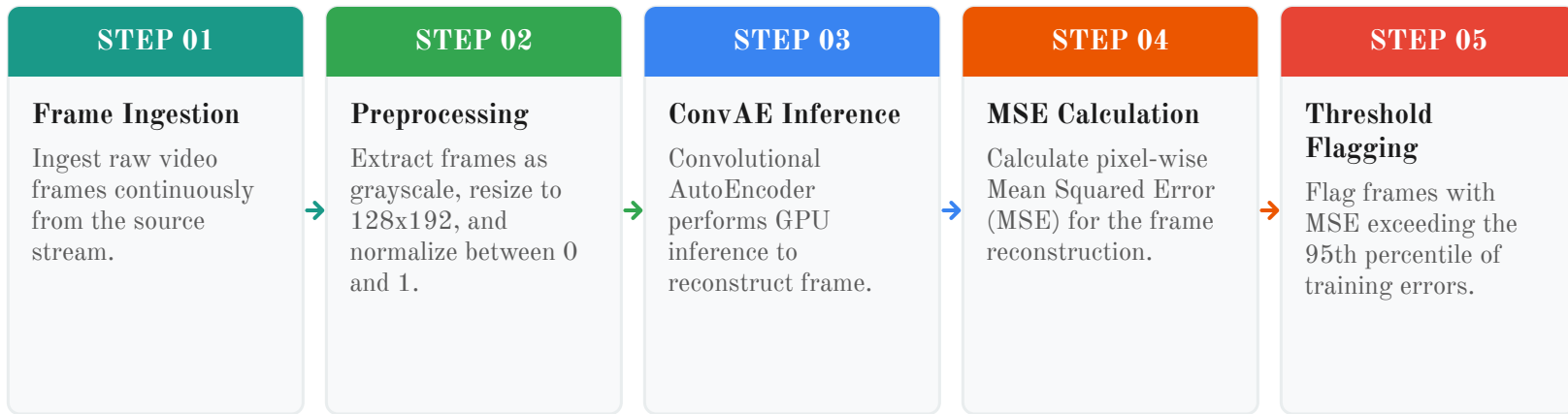
Dataset Sources & Frame Volume

- **Source Video:** Real outdoor CCTV surveillance footage recorded at **10 frames per second**.
- **Training Set:** **16 training clips** featuring normal behavior (**2,550 normal training frames**).
- **Test Set:** **12 test clips** containing normal behavior alongside anomalies.
- **Test Volume:** Total of **2,010 test frames**, of which **82.0%** contain anomalies (bicyclists, skaters, and carts).

Preprocessing & Evaluation Setup

- **Frame Properties:** Raw grayscale video frames resized to **128x192**.
- **Learning Paradigm:** Semi-supervised approach utilizing **pixel-level ground-truth masks** for test frames.
- **Anomaly Threshold:** Established at the **95th percentile** of training reconstruction errors.
- **Threshold Value:** Fixed precisely at **0.000734**.

Video Anomaly Detection Pipeline



Core Insight

High reconstruction error indicates the scene deviates from the learned normal distribution, failing to reconstruct unseen anomalies.

Convolutional AutoEncoder Baseline

Architecture Specifications

UNSUPERVISED ONE-CLASS MODEL

- **Input Resolution:** 128x192 frames
- **Encoder:** 4 convolutional layers using Stride-2, BatchNorm, and LeakyReLU
- **Bottleneck Layer:** 16x24x128 feature map
- **Decoder:** Mirrored structure reconstructing the original input frame
- **Total Parameters:** 257,761 trainable parameters
- **Loss Function:** Pixel-wise MSE loss (no anomaly labels required)

Training Parameters

- **Hardware:** RTX 4050 Laptop GPU
- **Training Time:** 143 seconds (30 epochs)
- **Batch Size & Optimizer:** Size 64 with Adam
- **Learning Rate:** CosineAnnealingLR

Modeling Rationale

Prioritized establishing a lightweight, domain-specific baseline before evaluating computationally expensive architectures like **PatchCore** or **DRAEM**.

Model Training & Loss Convergence

TRAINING HARDWARE & TIME

143 Seconds Total (RTX 4050 GPU)

CONVERGENCE DURATION

30 Epochs (Rapid Stabilization)

EPOCH 1

0.007710

Initial Loss
Training Start

EPOCH 5

0.000925

Significant Drop
Rapid Decline

EPOCH 15

0.000447

Continued Loss Reduction
Fine-Tuning

EPOCH 30

0.000367

Stabilized Loss
FINAL BASELINE

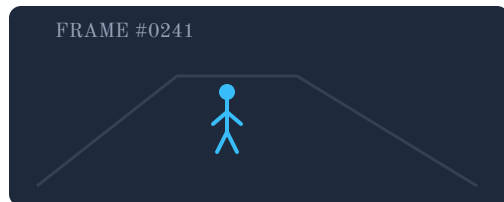
RECONSTRUCTION OUTCOME

As a result of this training process, the model successfully learns to reconstruct normal pedestrian walkway scenes with a highly consistent Mean Squared Error of approximately **0.000367**, establishing a strong baseline for normal behavior.

Visual & Pixel-Wise Error Heatmap Analysis

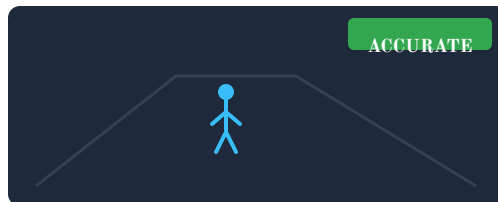
1. Raw Input Frame

NORMAL FRAME



2. ConvAE Reconstruction

ACCURATE



3. Pixel-wise Error Heatmap

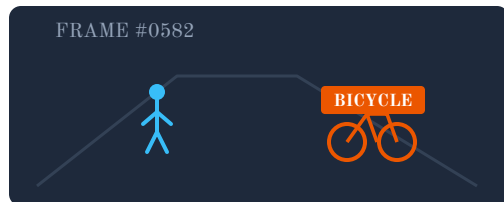
LOW / UNIFORM ERROR

MSE Score: 0.012 (Below Threshold)

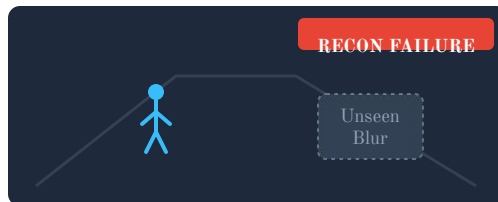


For **normal frames**, the reconstruction is nearly identical to the original, resulting in a low, uniform error distribution across the heatmap.

ANOMALY FRAME



RECON FAILURE



HOTSPOT DETECTED

MSE Score: 0.185 (SPIKE > THRESHOLD)



Conversely, **anomaly frames** reveal bright hotspots in the error map exactly where unexpected objects (e.g., bicycle or skater) appear.

Detection Mechanism: Because the model cannot reconstruct unseen objects, the Mean Squared Error (MSE) score spikes above the established threshold, successfully flagging the frame as anomalous.

Example Anomaly Score Timeline

Scope & Data Size

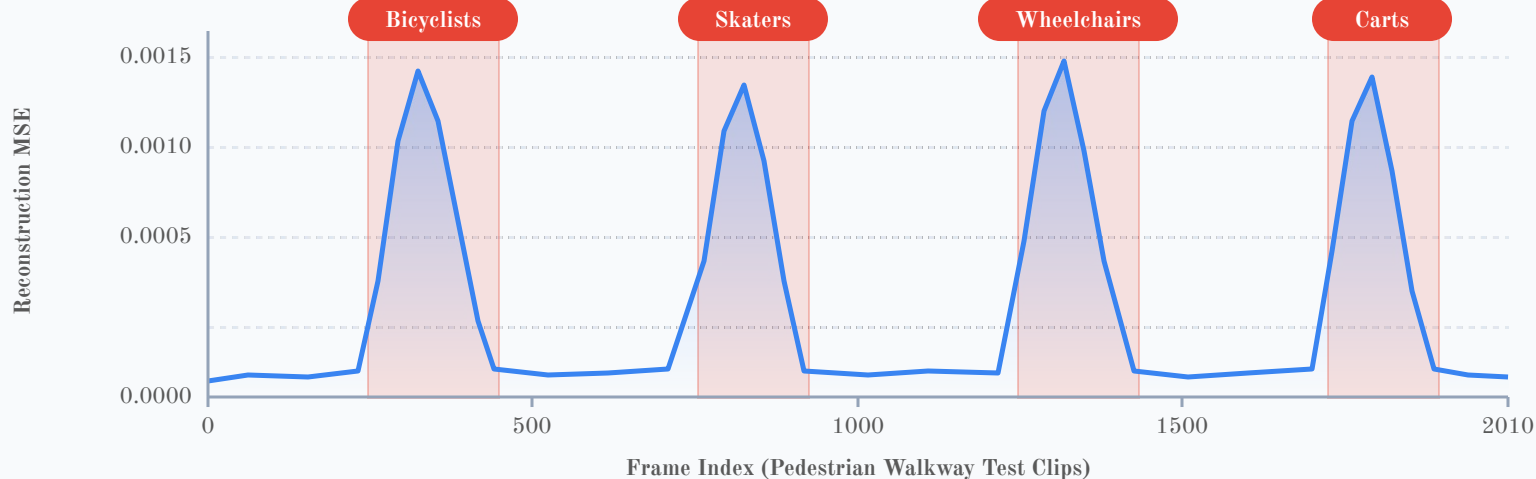
2,010 Test Frames across 12 Clips

Target Metric

Per-Frame Reconstruction MSE

Detection Threshold

Fixed Cutoff = 0.000734



Ground-Truth Anomaly Window

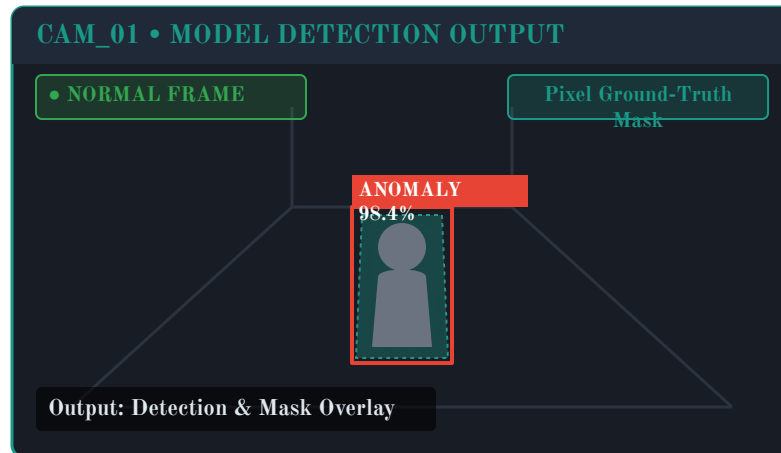
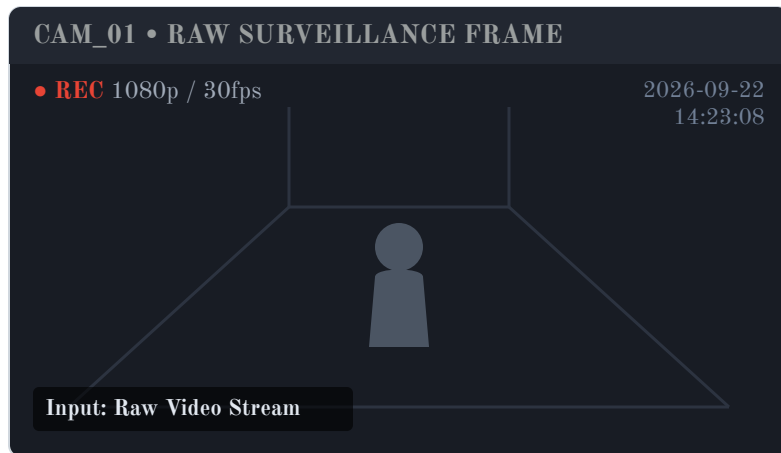


Per-Frame Reconstruction MSE



Detection Threshold (0.000734) — High Responsiveness

Detection Interface & Visual Layout



REAL-TIME PREDICTION EVALUATION BADGES (VS. PIXEL-LEVEL GROUND-TRUTH MASKS)

True Positive (TP)

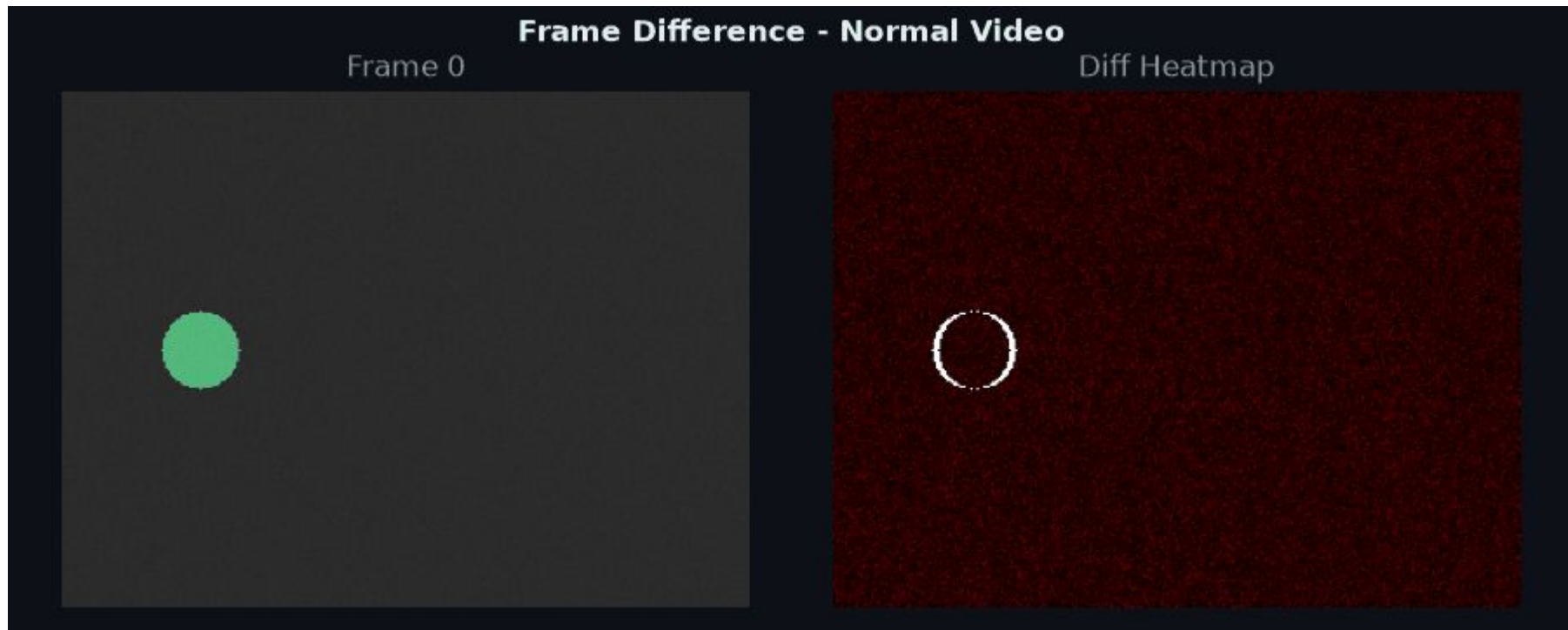
False Positive (FP)

False Negative (FN)

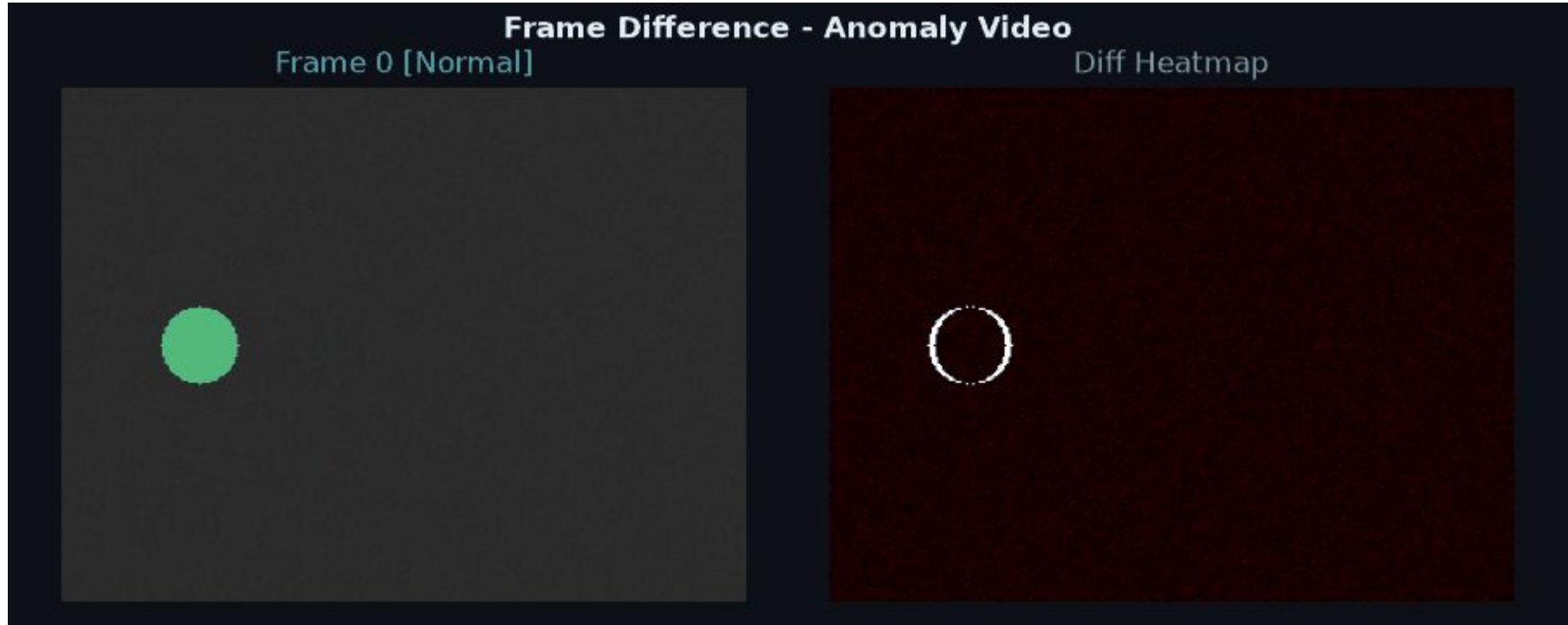
True Negative (TN)

- **Red Bounding Box:** Immediate visual indicator flagging an anomaly frame detected by the model.
- **Green Text / Status:** Visual indicator denoting a normal, non-anomalous surveillance frame.

Detection Interface & Visual Layout (Normal Video)



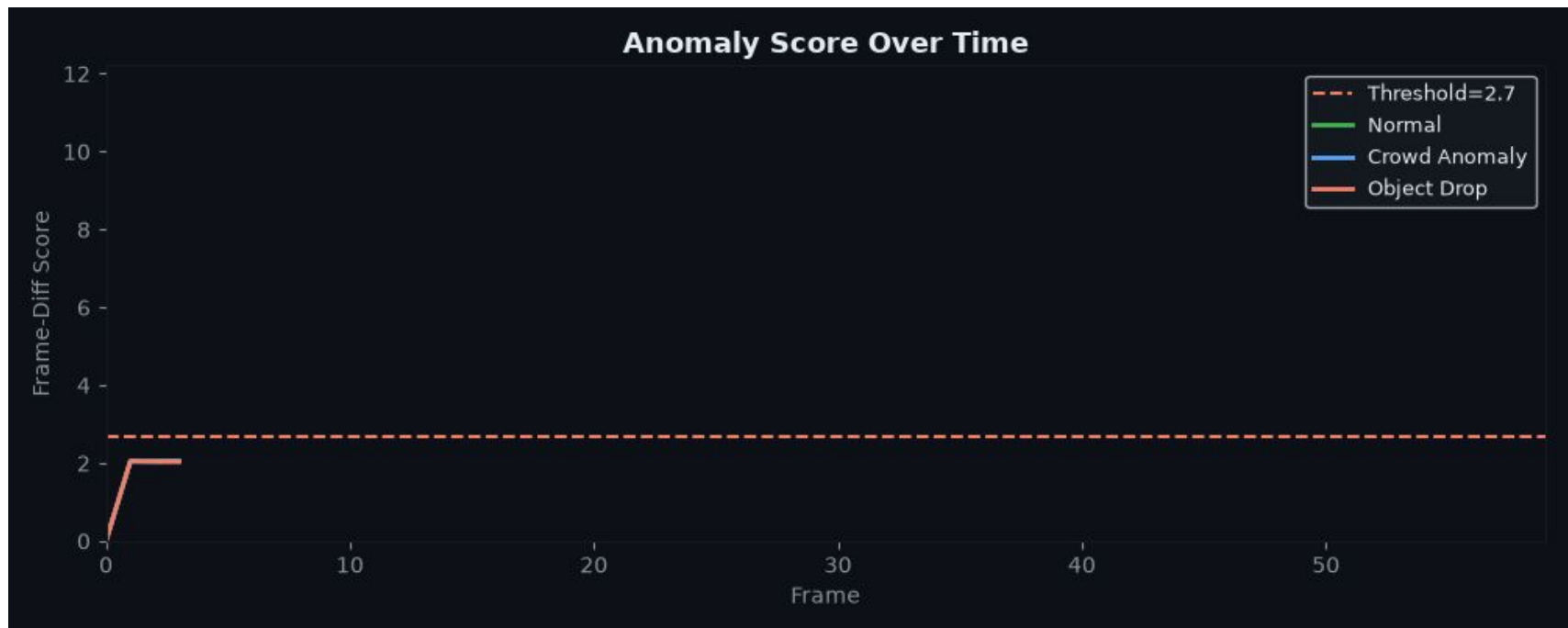
Detection Interface & Visual Layout (Anomalous Video)



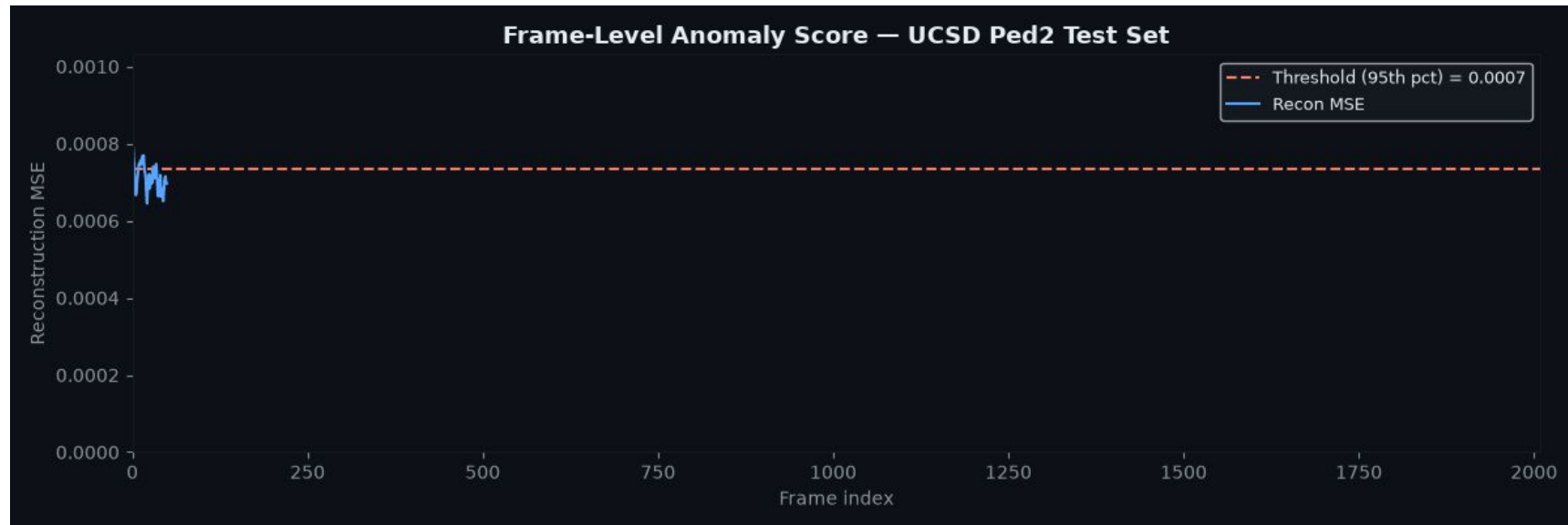
Detection Interface & Visual Layout (on actual dataset)



Anomaly score over time (on actual dataset)



Frame Level anomaly score over time (on actual dataset)



Detections (on actual dataset)

UCSD Ped2 — Frame 0 | GT: Normal | FP

Raw CCTV Frame



ConvAE Detection



Evaluation on UCSD Ped2 Test Set

Confusion Matrix

	Actual Positive	Actual Negative
Pred. Positive	358 True Positive (TP)	28 False Positive (FP)
Pred. Negative	1290 False Negative (FN)	334 True Negative (TN)

Production Focus: High precision was explicitly prioritized for deployment to maintain analyst trust and minimize false alarm fatigue.

Metric Selection: AUC-ROC was chosen as a standard threshold-free evaluation metric across operating points.

Key Metric Summary

Precision (Prioritized)

0.927

Recall

0.217

F1 Score

0.352

AUC-ROC Metric

0.6428

Benchmark Context (SOTA Comparison)

While the current model achieves an AUC-ROC of **0.6428**, state-of-the-art (SOTA) models such as **PatchCore** attain approximately **0.95** on similar evaluation benchmarks.

Evaluation & Deployment Strategy

Class Imbalance

Severe Imbalance

Test set features severe class imbalance, with 82% of frames being anomalous.

One-Class Model

Trained exclusively on normal data, making oversampling techniques like SMOTE unnecessary.

Threshold Control

The precision-recall tradeoff is strictly controlled by the threshold percentile.

Streaming Latency

< 5 ms

Online per-frame GPU latency

Latency Budget

Operates under 5 milliseconds on GPU, placing it well within the 33-millisecond budget.

Real-Time Execution

Designed for continuous online per-frame streaming processing.

Edge Efficiency

2 MB

Lightweight model checkpoint

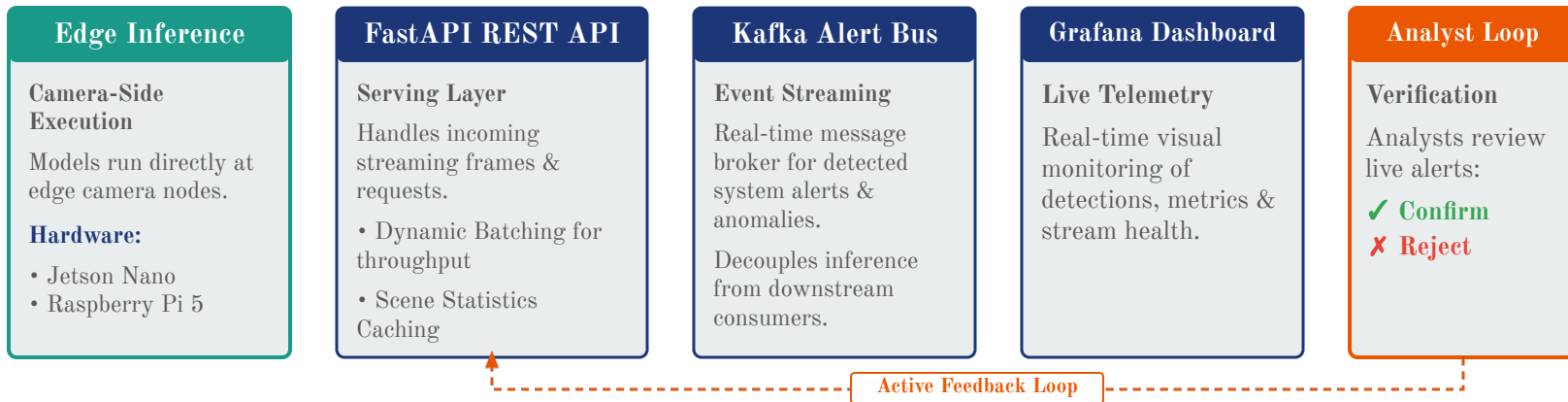
Edge Deployment

Model checkpoint easily fits on edge devices like Jetson Nano.

Future Efficiency Levers

- Batching & FP16 inference
- TensorRT export
- INT8 quantization

Proposed System Deployment Architecture



Serving Infrastructure & Latency Budget Performance

33 ms

Latency Budget

Maximum threshold per frame for real-time compliance

< 5 ms

GPU Inference

Operates significantly well within time budget

~30 ms

CPU Fallback

Maintains target rate per frame on edge CPU

Optimizations:

- **Batching:** High throughput handling
- **Caching:** Real-time scene statistics
- **Reliability:** Robust CPU fallback

Baseline System

Results & Analysis

0.6428

AUC-ROC

0.927

Precision

< 5 ms

Inference / frame

Real Results

Achieved strong precision using **zero anomaly training labels**, running at real-time speeds under 5ms per frame.

Root Cause Analysis

Low recall (**0.217**) stems from a conservative threshold and a single-frame architecture that misses temporal motion anomalies.

Next Steps & Lessons

Architectural Redo Steps

- Explore **3D Convolutional AutoEncoders** operating over 8-frame clips.
- Integrate **PatchCore** for enhanced patch-level feature extraction.
- Incorporate **optical flow channels** to explicitly capture dynamic motion.

Key Lesson Learned

Shipping a one-class unsupervised baseline fast earns **analyst trust through high precision**, establishing a solid foundation while recall is iteratively improved using temporal models.

Thank you!